

Predicting Showjumping Outcomes with Self-Supervised Video Representations: Type-Conditioned Approach Quality from Unlabeled Footage

Anonymous CVPR submission

Paper ID 0000

Abstract

001 *Whether a showjumping horse clears a fence or knocks a*
002 *rail is a motion-analysis problem, but predicting it from*
003 *broadcast video risks learning which arena a clip came*
004 *from rather than how the horse approached. We study type-*
005 *conditioned approach quality at a fixed 1.50 m height, using*
006 *a self-supervised R(2+1)D-18 video encoder pretrained on*
007 *unlabeled clips and fine-tuned with small heads. Our cen-*
008 *tral finding from earlier work was that the encoder's domi-*
009 *nant axis of variation was source venue instead of approach*
010 *quality. We attack this with a venue-balanced sampler and*
011 *a domain-adversarial objective intended to suppress venue*
012 *identity, and we evaluate under an honest leave-one-venue-*
013 *out protocol whose folds never share a backdrop between*
014 *train and test. Our central finding is that although the ad-*
015 *versarial domain loss converges to chance, a linear probe*
016 *still recovers venue at 85%, and a t-SNE that looks fully de-*
017 *biased disagrees with the probe; a converged adversarial*
018 *loss is not invariance. This residual venue signal explains*
019 *that cross-venue outcome F1 sits at the single-class floor,*
020 *and the proposed SSL label-efficiency advantage, though*
021 *present as a low-budget trend, is within seed noise.*

022 1. Introduction

023 Whether a horse jumps clean or knocks a rail is decided
024 by motion: stride rhythm, approach speed, and the takeoff
025 distance d determine the outcome. Predicting outcome di-
026 rectly from video, however, conflates how well the horse is
027 approaching and how hard the fence is. A vertical and an
028 oxer of the same height pose different problems, so a model
029 trained on mixed fences can shortcut by learning fence type
030 instead of approach quality. We fix the height at 1.50 m and
031 study approach quality conditioned on obstacle type (verti-
032 cal vs. oxer), taking takeoff distance d as a geometric target
033 recovered from video.

Motivation Per-jump outcome labels are expensive and
require domain expertise, while unlabeled 1.50 m footage is
abundant publicly. This is exactly the regime where self-
supervised learning (SSL) should be used. We pretrain
a video encoder on unlabeled clips, then fine-tune small
heads on a modest labeled set. We test the hypothesis that
SSL-pretrained representations match from-scratch super-
vised accuracy with half the labels, and beat ImageNet pre-
training with fewer still. Beyond equestrian sport, the set-
ting is a clean example of learning behavior representations
from broadcast video where the recording environment is
strongly correlated with, but causally irrelevant to, the label.
That correlation is the central difficulty of this project. In
our earlier pipeline, t-SNE of the encoder embeddings clus-
tered cleanly by source broadcast — arena, lighting, foot-
ing, and crowd — rather than by approach quality. Because
a stratified split lets the same venue appear in both training
and test, a model can score well by memorizing a venue.
We therefore treat venue as a confound to be measured and
removed.

Contributions

1. A diagnosis of domain-adversarial debiasing on broad-
cast video: the DANN domain loss converges to chance,
yet a linear probe still recovers venue at 85% and a t-
SNE looks fully debiased. A converged adversarial loss
is not invariance, and is only revealed by a direct probe.
2. A leave-one-venue-out protocol (each venue is a distinct
broadcast, so the held-out arena is wholly unseen) plus
the venue probe and sample-efficiency study that surface
the above; under it, the proposed SSL label-efficiency
gain is at best a within-noise low-budget trend.
3. An end-to-end pipeline from raw broadcasts to (type, d ,
outcome) labels (a geometric takeoff-distance estimator
calibrated by the standard 3.5 m pole, knockdown min-
ing, and a cross-venue labeled set).

2. Related Work

Self-supervised video representation learning Contrastive instance discrimination (SimCLR [2], MoCo [4]) pulls augmented views of a sample together and pushes other samples apart via the InfoNCE loss [14]. Because appearance-only contrast can ignore motion, video methods add temporal pretext tasks (order verification [10], clip-order prediction [17], pace prediction [16]). We combine InfoNCE with a 6-way temporal-order task on an R(2+1)D-18 backbone [13]. Unlike these methods, evaluated on action-recognition benchmarks where domain is balanced across classes, our footage couples domain (venue) tightly to the recording, so contrastive negatives are venue-dominated unless explicitly controlled.

Domain shift and adversarial debiasing Domain-adversarial training [3] uses a gradient-reversal layer to make features uninformative about domain while staying predictive of the task. Normally applied to cross-dataset adaptation, we repurpose “domain” as the source broadcast within a single task, stripping venue cues (arena, crowd, graphics) that would otherwise act as an outcome shortcut, an instance of the broader problem that vision models exploit backgrounds rather than the object of interest.

Equestrian motion analysis Computer vision for sports centers on tracking, pose, and event detection [12]. Quantitative equine-jumping analysis is mature but relies on self-collected, controlled data: Santamaría et al. [11] track skin markers to measure jumping technique (including the fence-base-to-limb take-off distance we also estimate) and Barrey and Galloux [1] fit an accelerometer belt to eight horses; markerless DeepLabCut [8] pipelines and horse datasets such as Horse-10 [9] and the marker-validated PFERD [7] (5 horses, ten cameras, including jumping) now supply ground-truth kinematics. All are similar in that they examine a few horses recorded by the authors with fixed cameras or worn sensors. We instead learn from pre-existing broadcast footage (three Grand Prix competitions). Uncontrolled video carries a venue signal controlled capture never exposes. To our knowledge this is the first work to learn self-supervised representations of jumping from public competition video, recover takeoff distance geometrically from broadcast frames, and predict jump outcome under an explicit cross-venue protocol. We use YOLOv8 [5] for detection and t-SNE [15] for embedding visualization.

3. Methodology

3.1. Data and Preprocessing

Footage and segmentation We scraped one full 1.50 m Grand Prix broadcast from each of three venues (Tryon In-

ternational, Club de Campo Villa de Madrid, and Wellington International) and cut them into 2-second clips at 16 fps, 320×320 . A YOLOv8 [5] horse-presence filter (keep clips with a horse visible in $\geq 40\%$ of sampled frames) removes crowd shots, course walks, slow-motion replays, and interviews, leaving 2,142 candidate jumping clips, one broadcast per venue. An earlier version of the corpus drew four of its six broadcasts from Wellington, over-representing that single arena and skewing both the venue-balanced sampler and the domain-adversarial head toward it; we deliberately trimmed the set to one broadcast per venue so the three venues contribute comparable footage and leave-one-venue-out is symmetric across them. The trade-off is that, with one competition per venue, venue, competition, course, and camera setup are confounding.

Outcome labels and knockdown mining Clean jumps dominate broadcasts, so naive sampling yields almost no negatives. At demo day the labeled set held 110 outcomes (101 “clean,” 9 “knockdown”) and the classifier collapsed to always predicting “clean” (macro-F1 ≈ 0.47 , the single-class baseline). We therefore mined knockdowns by scrubbing to replay/commentary-flagged rails and cutting tight clips around the fault, growing the set to 346 clips across all three venues — 256 clean and 90 knockdown (2.84:1; madrid 118/15, tryon 65/40, wellington 73/35) — a $3\times$ scale-up over the demo-day 110 that also sharply improves balance. Because refusals are too rare to label reliably at this height, the outcome task reduces to binary clean-vs-knockdown (a deviation from the proposed three-class target; see §5). The set is still imbalanced, so we report macro-F1 as the headline metric.

Geometric takeoff distance Pretrained COCO YOLOv8 (the horse class) detects the horse every other frame. The takeoff frame is the sharpest downward-to-upward reversal of the horse-box bottom edge in the clip’s second half, a proxy for the hoof leaving the ground. On the subset where the takeoff actually falls inside the clip window ($n=27$) the detector localizes it to a mean absolute error of 3.3 frames (67% within ± 2); across all 77 annotated clips the error rises to 7.0 frames because roughly two-thirds have the takeoff cut at the segment boundary — a footage/segmentation limit. The standard 3.5 m top pole gives meters-per-pixel $mpp = 3.5/w_{fence}$, and d is the ground distance from the takeoff-frame hoof line to the fence base, scaled by mpp (Fig. 1). The fence boxes are 338 hand-annotated rectangles across the three videos, typed by pole count (215 single-plane verticals, 123 two-pole oxers); we additionally fine-tuned a YOLOv8 fence detector on them to recover the proposal’s automated detection path.



Figure 1. **Detection and geometry.** *Top*: takeoff-frame localization (box-edge reversal, approach→suspension→landing); MAE 3.3 frames (67% within ± 2) on the $n=27$ in-window clips (see text). *Bottom*: horse box (green), fence box (orange), and geometric d at the detected takeoff frame; bottom-right is the oblique-angle failure mode that inflates d .

3.2. Self-Supervised Encoder

The encoder is a torchvision R(2+1)D-18 trunk [13] with a 2-layer MLP projector to 128 dimensions. Three objectives are trained jointly:

- **InfoNCE** [14] on two photometric- and flip-augmented views of a clip ($\tau=0.1$), with the rest of the batch as negatives.
- **6-way temporal-order** classification of the permutation applied to three disjoint 4-frame sub-clips, weight $\lambda_{\text{order}}=0.3$.
- **Venue-adversarial** prediction of source venue behind a gradient-reversal layer [3], weight $\lambda_{\text{domain}}=0.3$, pushing the encoder to discard venue-identifying cues.

We optimize with AdamW (lr 10^{-3} , weight decay 10^{-4} , cosine schedule), batch 32, from Kinetics-400 [6] initialization. From-scratch SSL collapsed (nearest-neighbor cosine ≈ 0.97) at this data scale. A venue-balanced sampler equalizes venue representation per batch so contrastive negatives are not dominated by a single broadcast.

3.3. Downstream Heads and Baselines

On the frozen or fine-tuned encoder, we train a type-conditioned outcome classifier and a d regressor. We compare against three baselines: from-scratch supervised R(2+1)D-18, an ImageNet-pretrained 2D-CNN, and a type-only logistic regression. The last shows that if fence type alone predicts outcome well then the task is not measuring approach quality.

3.4. Honest Cross-Venue Evaluation

A stratified split leaks venue; because each venue is a single broadcast, the same arena, crowd, and on-screen graphics would appear in both train and test, letting a model score well by memorizing venue characteristics. With

one competition per venue, venue, course, and backdrop are perfectly confounded, so we cannot construct a same-venue/different-competition control. We therefore evaluate strictly leave-one-venue-out: hold out one venue’s clips entirely, train on the other two, and repeat for all three folds. The held-out arena is never seen in training, making this a strict test of cross-venue generalization. We quantify residual venue leakage directly with a venue linear probe on frozen embeddings (chance = 13). Sample efficiency is reported under the same held-out-venue split at $n \in \{25, 50, 100\}$ with three seeds, so the SSL-vs-baselines comparison is never inflated by venue leakage.

4. Results and Analysis

Our strongest and most reproducible finding is a negative one about the venue confound; the proposal’s headline sample-efficiency comparison, by contrast, is inconclusive at this data scale. We present them in that order.

4.1. The venue confound survives adversarial debiasing

This is the paper’s central result. During pretraining the adversarial domain loss converged to its three-class chance floor ($L_d \approx \log 3 \approx 1.10$). The venue head could no longer tell broadcasts apart, which is what a successful DANN looks like from the inside. Yet a post-hoc linear probe trained on the frozen embeddings recovers venue at 0.845 ± 0.028 (chance 0.333, majority 0.389; $n=2142$). The features retain venue information such that the DANN objective was satisfied without achieving invariance.

The t-SNE in Fig. 2 sharpens this point. Both the DANN encoder and an info-only ($\lambda_{\text{domain}}=0$) ablation are heavily venue-mixed in 2D (an improvement from the milestone), so the projection alone would read as “confound removed.” It is not: the same DANN embeddings are 85% venue-decodable. The 2D appearance and the linear decodability give different results, but the probe is quite decisive.

4.2. Cross-venue outcome prediction is at the floor

Because each venue is a single broadcast, there is no within-venue control; we evaluate strictly leave-one-venue-out. Test A (Fig. 3) yields mean macro-F1 0.46 (tryon 0.52, wellington 0.45, madrid 0.43) at mean accuracy 0.58. This sits at the single-class floor (~ 0.47), with only tryon clearly above it. Approach-quality signal does not transfer to an unseen venue at three venues, directly consistent with §4.1. If the embeddings still encode venue, a held-out venue is effectively out of distribution. The fold-to-fold spread is large, reflecting how strongly each held-out venue’s class balance and appearance dominate at this scale.

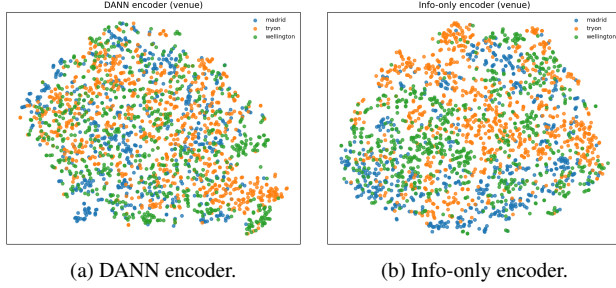


Figure 2. **Embeddings colored by venue:** DANN (left) vs. info-only (right). Both look venue-mixed in 2D, so the projection alone reads as debiased, yet the probe recovers venue at 0.845 from the DANN features. Appearance and decodability are unagreeable but the probe is decisive.

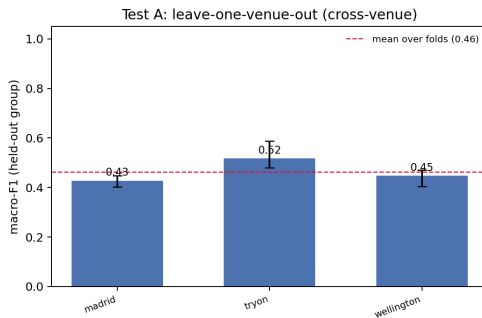


Figure 3. **Leave-one-venue-out (Test A).** Per-fold macro-F1 (mean, min/max over seeds): tryon 0.52, wellington 0.45, madrid 0.43; mean 0.46. Only tryon clears the ~ 0.47 single-class floor; with three folds, variance is high.

4.3. Sample efficiency: inconclusive

The proposal’s central quantitative test is whether SSL cuts the label budget. Our result is weak. Under a rotating held-out venue (Fig. 4, Table 1) every method sits in a narrow 0.35–0.56 macro-F1 band around the ~ 0.47 single-class floor, and the seed-to-seed spread dwarfs the between-method gaps. SSL at $n=25$ alone spans 0.44–0.61 across its three seeds. With three folds and $n_{\text{val}} \approx 107$, these differences are not statistically separable. The most we can say is that SSL is never the worst method and trends best in the low-label regime; we do not claim a clean ranking or a specific “ $N \times$ label saving.” The ordering is also venue-sensitive (on a fixed madrid hold-out, (§12b) ImageNet-2D leads instead).

4.4. The minority class is barely detected

The task’s real target is near-undetected at this scale. On the confusion fold (Fig. 5, $n_{\text{val}}=27$, two knockdowns) the classifier predicts both knockdowns as clean – knockdown recall 0, i.e. it collapses to a near-clean predictor. The larger held-out-venue sweep shows weak but non-zero knock-

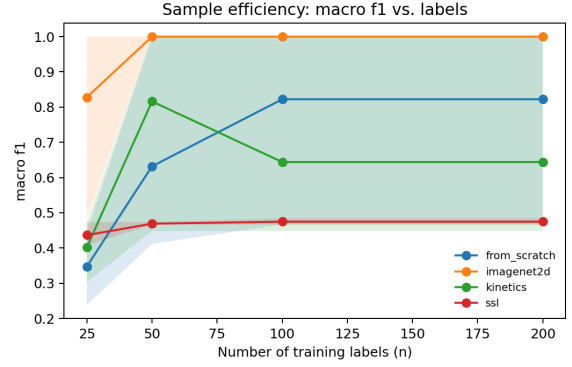


Figure 4. **Sample efficiency under a held-out venue.** Macro-F1 vs. labeled clips $n \in \{25, 50, 100, 200\}$ (rotating held-out venue). All methods overlap around the single-class floor (dashed); between-method differences are within seed variance.

Method (held-out venue)	$n=25$	$n=50$	$n=100$	$n=200$
SSL (ours)	0.51	0.50	0.54	0.56
ImageNet-2D CNN	0.36	0.53	0.50	0.54
Kinetics-only	0.43	0.37	0.45	0.52
From-scratch supervised	0.35	0.49	0.39	0.48

Table 1. **Sample-efficiency macro-F1** (seed means, rotating held-out venue). All values sit within ~ 0.1 of the 0.47 single-class floor; differences are within seed spread and should not be read as a ranking (see §Sample efficiency).

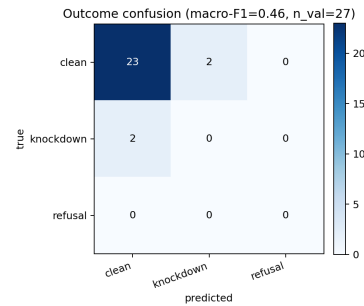


Figure 5. **Confusion matrix**, fine-tuned SSL classifier (macro-F1 0.46, $n_{\text{val}}=27$). Both knockdowns are predicted clean — knockdown recall 0 on this fold. Minority detection is unreliable; the larger sweep (Fig. 4) is the more trustworthy signal.

down signal (macro-F1 above the floor on some folds), but no evaluation shows reliable knockdown detection. We report macro-F1 rather than accuracy precisely because accuracy hides this; the binary clean-vs-knockdown task is only weakly learned.

Geometric d The geometric estimator yields physically plausible takeoff distances (1–3 m; Fig. 1). Learned d -

272 regression, however, is not useful. Across the sweep the
273 regressor reaches a d -MAE of ≈ 1.2 m on that 1–3 m range,
274 comparable to or worse than simply predicting the mean, so
275 we report it as a negative result and defer a usable regressor
276 to future work.

277 4.5. Discussion and Limitations

278 **The venue confound.** Every downstream weakness
279 above traces back to §4.1. With venue still 85% decodable,
280 an unseen venue is out of distribution, so cross-venue F1
281 sits at the floor and label-efficiency differences vanish into
282 noise. Stronger invariance (a larger venue penalty, a deeper
283 adversary, or more than three venues so the adversary can-
284 not memorize) is the obvious next step.

285 **Type conditioning gives no measurable benefit.** The ti-
286 tle frames type-conditioned approach quality, but a held-
287 out-venue ablation ($n_{\text{val}} \approx 107$, 3 seeds) does not sup-
288 port the conditioning. The type-conditioned head reaches
289 macro-F1 0.55 versus 0.58 for the unconditioned one, if
290 anything slightly worse, and within seed noise either way.
291 Type conditioning is thus a modeling choice our data does
292 not validate. Whether it helps at larger scale, or with a
293 learned rather than pole-count fence type, is uncertain.

294 **Class imbalance and small folds.** With a clean-
295 dominated, refusal-free set, accuracy is misleading and
296 macro-F1 is most important. Leave-one-venue-out gives
297 one venue per fold ($n_{\text{val}} = \text{madrid } 133, \text{ tryon } 105, \text{ welling-}$
298 $\text{ton } 108$) and the confusion fold is smaller (27); three folds
299 and three seeds remain noisy, which is itself a limitation on
300 every claim above.

301 **Geometric d depends on a fragile takeoff frame.** The
302 pole-scaled estimator inflates d when the camera is off-axis
303 (Fig. 1), and d inherits the takeoff-frame error: reliable only
304 on the $\sim$ one-third of clips whose takeoff falls inside the 2-
305 second window (MAE 3.3 frames), and unreliable on the
306 rest, where it is cut at the boundary. Side-view filtering and
307 wider windows would trade pool size for accuracy.

308 5. Conclusion

309 We studied type-conditioned showjumping outcome pre-
310 diction as a self-supervised video-representation problem
311 whose central obstacle is a venue confound such that broad-
312 cast environment correlates strongly with, but is causally
313 irrelevant to, the label. We attacked it with a venue-
314 balanced sampler and a domain-adversarial objective, and
315 evaluated under a strict leave-one-venue-out protocol. The
316 most important finding is that the domain-adversarial de-
317 biasing did not achieve invariance, although its loss con-
318 verged to chance, venue stayed $\sim 85\%$ linearly decodable,

and a t-SNE that looks debiased disagrees with the probe. 319
Cross-venue outcome F1 sits at the single-class floor, the 320
proposal’s SSL label-efficiency advantage appears only as a 321
low-budget trend within seed noise rather than a separable 322
win, and the minority class (rails down) is only weakly de- 323
tected at this scale. This confound caps how far any of this 324
work transfers to accuracy. 325

Future work Larger and more balanced knockdown la- 326
bels, recovery of the three-class outcome (including re- 327
fusals), isolating the individual pretext tasks (InfoNCE-only 328
vs. temporal-order-only, which we did not run, only the 329
type-conditioning ablation, which showed no benefit), re- 330
instating d -regression with side-view filtering, and the de- 331
ferred future-frame prediction task once the confound is 332
controlled. 333

Key takeaway A converged adversarial loss is not invari- 334
ance. Our venue head reached its chance floor while a linear 335
probe still read venue at 85%. Evaluating self-supervised 336
video representations under domain shift therefore demands 337
both a leakage-free split and a direct probe of the nuisance 338
variable. 339

513 6. Individual Contributions

This is a solo project. I designed and implemented the 514
full pipeline (scraping, segmentation, horse/fence detec- 515
tion, geometric d , knockdown mining, SSL pretraining with 516
the venue-adversarial head, downstream heads, baselines, 517
and the cross-venue evaluation), and ran all experiments. 518
Claude Code was used to help write the in-Colab annota- 519
tor and scraping script, evaluation/visualization scaffolding, 520
and code documentation. Codebase: <https://github.com/bballhaus/showjumping-ssl>. 521

522 References

- [1] E. Barrey and P. Galloux. Analysis of the equine jumping 523
technique by accelerometry. *Equine Veterinary Journal Sup-* 524
plement, (23):45–49, 1997. 2 525
- [2] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Ge- 526
offrey Hinton. A simple framework for contrastive learn- 527
ing of visual representations. In *International Conference on* 528
Machine Learning (ICML), 2020. 2 529
- [3] Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pas- 530
cal Germain, Hugo Larochelle, François Laviolette, Mario 531
Marchand, and Victor Lempitsky. Domain-adversarial train- 532
ing of neural networks. *Journal of Machine Learning Re-* 533
search (JMLR), 17(59):1–35, 2016. 2, 3 534
- [4] Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross 535
Girshick. Momentum contrast for unsupervised visual rep- 536
resentation learning. In *IEEE/CVF Conference on Computer* 537
Vision and Pattern Recognition (CVPR), 2020. 2 538

- [5] Glenn Jocher, Ayush Chaurasia, and Jing Qiu. Ultralytics yolov8. <https://github.com/ultralytics/ultralytics>, 2023. 2
- [6] Will Kay, Joao Carreira, Karen Simonyan, Brian Zhang, Chloe Hillier, Sudheendra Vijayanarasimhan, Fabio Viola, Tim Green, Trevor Back, Paul Natsev, Mustafa Suleyman, and Andrew Zisserman. The kinetics human action video dataset. *arXiv preprint arXiv:1705.06950*, 2017. 3
- [7] Ci Li, Ylva Mellbin, Johanna Krogager, Senya Polikovsky, Martin Holmberg, Nima Ghorbani, Michael J. Black, Hedvig Kjellström, Silvia Zuffi, and Elin Hernlund. The poses for equine research dataset (PFERD). *Scientific Data*, 11(1):497, 2024. 2
- [8] Alexander Mathis, Pranav Mamidanna, Kevin M. Cury, Taiga Abe, Venkatesh N. Murthy, Mackenzie W. Mathis, and Matthias Bethge. DeepLabCut: Markerless pose estimation of user-defined body parts with deep learning. *Nature Neuroscience*, 21(9):1281–1289, 2018. 2
- [9] Alexander Mathis, Thomas Biasi, Steffen Schneider, Mert Yükekçönlü, Byron Rogers, Matthias Bethge, and Mackenzie W. Mathis. Pretraining boosts out-of-domain robustness for pose estimation. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2021. 2
- [10] Ishan Misra, C. Lawrence Zitnick, and Martial Hebert. Shuffle and learn: Unsupervised learning using temporal order verification. In *European Conference on Computer Vision (ECCV)*, 2016. 2
- [11] S. Santamaría, M. F. Bobbert, W. Back, A. Barneveld, and P. R. van Weeren. Effect of early training on the jumping technique of horses. *American Journal of Veterinary Research*, 66(3):418–424, 2005. 2
- [12] Graham Thomas, Rikke Gade, Thomas B. Moeslund, Peter Carr, and Adrian Hilton. Computer vision for sports: Current applications and research topics. *Computer Vision and Image Understanding (CVIU)*, 159:3–18, 2017. 2
- [13] Du Tran, Heng Wang, Lorenzo Torresani, Jamie Ray, Yann LeCun, and Manohar Paluri. A closer look at spatiotemporal convolutions for action recognition. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018. 2, 3
- [14] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018. 2, 3
- [15] Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-sne. *Journal of Machine Learning Research (JMLR)*, 9:2579–2605, 2008. 2
- [16] Jangliu Wang, Jianbo Jiao, and Yun-Hui Liu. Self-supervised video representation learning by pace prediction. In *European Conference on Computer Vision (ECCV)*, 2020. 2
- [17] Dejing Xu, Jun Xiao, Zhou Zhao, Jian Shao, Di Xie, and Yueting Zhuang. Self-supervised spatiotemporal learning via video clip order prediction. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019. 2